MICROFLUIDIC MODELING FOR FLUIDS USING TESLA MICROVALVES

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Introduction

This work presents an innovative approach to solving instrumentation challenges in ultra-high vacuum systems, with a focus on optimizing the geometry of a Tesla microvalve. Tesla valves are devices that enable one-way fluid or gas flow while restricting reverse flow. These valves hold significant potential in a variety of applications, from microfluidic systems to large-scale fluid control systems. Maintaining ultra-high vacuum and vacuum break-free environments is crucial in various scientific and some industrial applications, such as vacuum-based sample analysis using ultraviolet vacuum (UVX) or soft X-rays, as well as in clean surface analysis and differential pumping. Achieving this goal requires efficient control of airflow.

This study centers on the simulation and optimization of Tesla microvalves using Computational Fluid Dynamics (CFD) software, specifically OpenFOAM, to simulate flow behavior in various geometries. We employ machine learning techniques, particularly neural networks, to extract flow behavior patterns for different valve geometries. Additionally, we implement a genetic algorithm to find the optimal geometry for the Tesla microvalve that maximizes airflow control efficiency while maintaining a high vacuum environment. This multidisciplinary approach offers a significant contribution to addressing instrumentation challenges in vacuum chambers, with the potential to enhance efficiency in a wide range of scientific and industrial applications.

Methodology

- Developing microvalve geometries similar to Tesla's original design.
- Generating a high-quality mesh using Gmsh software.
- Setting up of the simulation using CFD in software OpenFOAM: fluid properties, initial conditions and boundary conditions.
- Conducting simulation for various cases with different geometric parameters.
- Developing a neural network model with geometric parameters as inputs and airflow as output.
- Using the neural network-generated function to determine the optimal geometry that minimizes airflow with genetic algorithm.

Modelling

The solver employs the PIMPLE algorithm, a combination of PISO (Pressure Implicit with Splitting of Operator) and SIMPLE (Semi-Implicit Method for Pressure Linked Equations). This iterative solver is employed to solve the continuity and momentum equations:

$$\nabla \cdot \vec{\mathbf{u}} = 0 \tag{1}$$

$$\frac{\partial \vec{\mathbf{u}}}{\partial t} + \vec{\mathbf{u}} \cdot (\nabla \times \vec{\mathbf{u}}) - \nabla \cdot (\nu \nabla \vec{\mathbf{u}}) = -\nabla p \tag{2}$$

where $\vec{\mathbf{u}}$ is velocity, ν is kinematic viscosity and p is kinematic pressure.

- Fluid properties: $\rho = 998$ kg/m^3, $\nu = 1.012\ 10^{-6}$
- Initial conditions: u = $\pm$ 0.1 m/s, $\pm$ 1.0 m/s (direct and reverse fluxes)
- Boundary conditions: zero gradient pressure, no slip walls

Optimization

Geometry

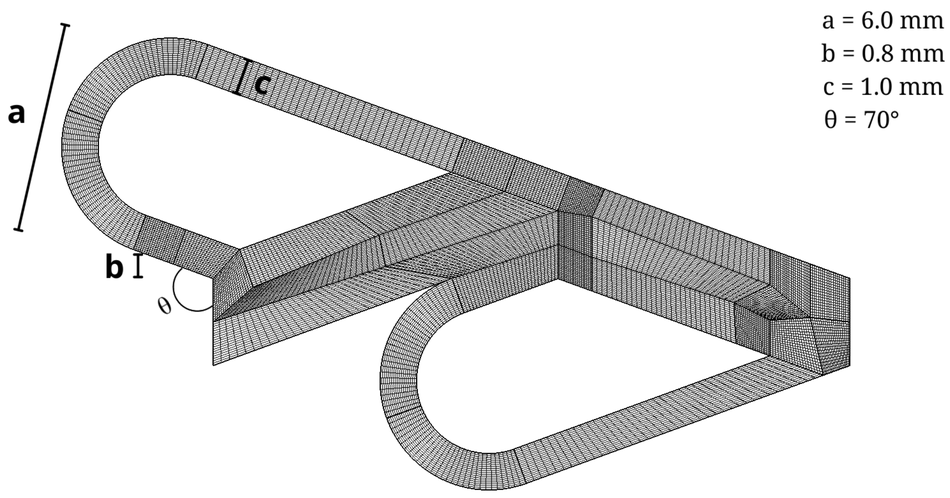


Figure 1. Geometric parameters and mesh.

Neural Network

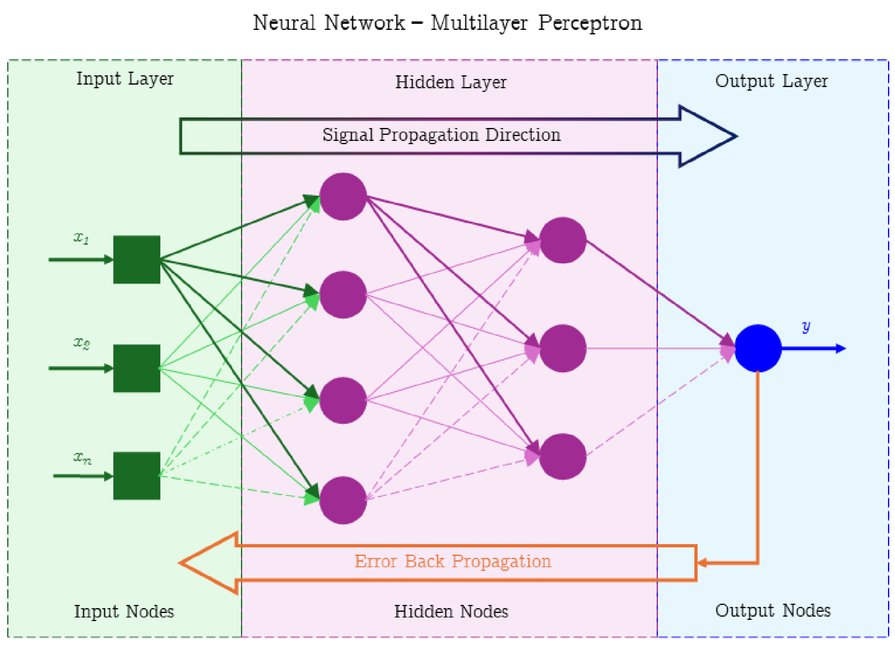


Figure 2. Neural network applied to generated data.

Results

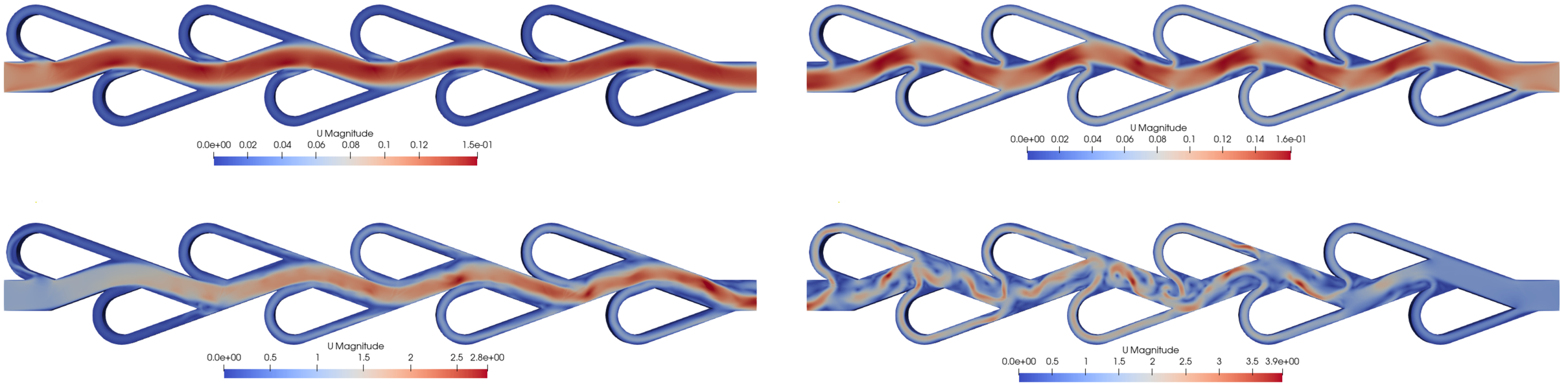


Figure 3. Velocity profile for (a) direct and (b) reverse fluxes for inlet velocities 0.1 and 1.0 m/s.

Future Work

- Vary geometry and input flux for fixed pressure gradient.
- Use a cluster to perform all simulations.
- Obtain the optimal geometry using machine learning (neural network and genetic algorithm).
- Construct the optimal microvalve and perform experiments to validate the simulations.

References

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